ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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C02-4

EXPERIMENT 9

Implementation of Magnetic-Field Boundary Conditions between Two Dielectric Media and Determination of Relative Permeability

Aim :

To implement magnetic-field boundary conditions between two media and determine the relative permeability of the unknown medium.

Test Case (i) – Magnetic Boundary Conditions and Determination of the Relative Permeability

Apply the boundary conditions H2t = H1t and B2n = B1n at the interface, determine the relative permeability of medium 2 from the measured normal H-field and compare it with the reference value.

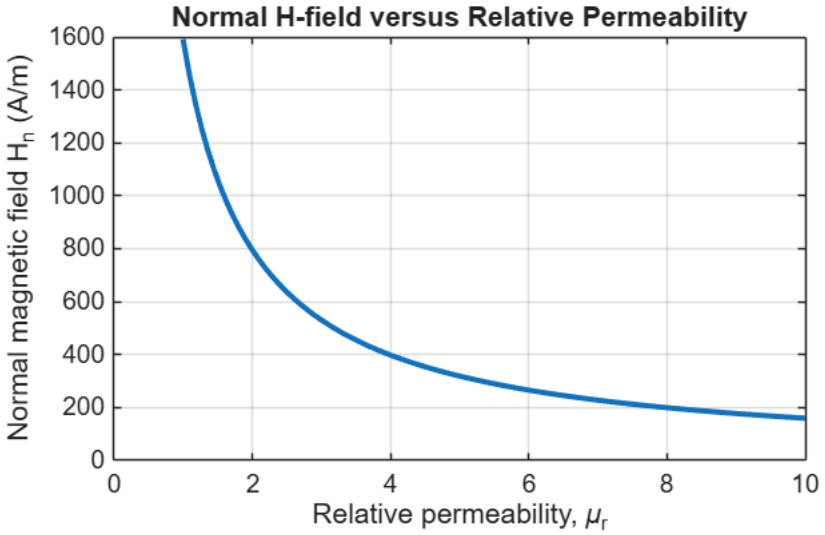
MATLAB Program:

```
clc;
clear;
close all;
%% Magnetic-Field Boundary Conditions between Two Media
mu0 = 4*pi*1e-7; % H/m
% Input data
mur1 = input('Enter relative permeability of medium 1: ');
H1t = input('Enter tangential H-field H1t (A/m): ');
B1n = input('Enter normal flux density B1n (T): ');
H2n_meas = input('Enter measured/assigned normal H-field H2n (A/m): ');
mur2_ref = input('Enter reference relative permeability of medium 2: ');
% Permeability of the two media
mu1 = mu0*mur1;
mu2_ref = mu0*mur2_ref;
% Boundary conditions for zero surface current
H2t = H1t;
B2n = B1n;
% Determine relative permeability from measured H2n
mur2_calc = B2n/(mu0*H2n_meas);
% Theoretical normal H-field using the reference permeability
H2n_theory = B2n/mu2_ref;
% Percentage deviation
dev = abs(mur2_calc-mur2_ref)/abs(mur2_ref)*100;
fprintf('\n--- Magnetic Boundary Condition Results ---\n');
fprintf('H2t (theoretical) = %.6f A/m\n',H2t);
fprintf('B2n (theoretical) = %.6e T\n',B2n);
```

```
fprintf('H2n (theoretical) = %.6f A/m\n',H2n_theory);
fprintf('Calculated mur2 = %.6f\n',mur2_calc);
fprintf('Reference mur2 = %.6f\n',mur2_ref);
fprintf('Percentage deviation = %.4f %%\n',dev);
% Parameter sweep: effect of relative permeability on normal H-field
mur = linspace(1,10,100);
Hn = B1n./(mu0*mur);
figure;
plot(mur,Hn,'LineWidth',2);
grid on;
xlabel('Relative permeability, \mu_r');
ylabel('Normal magnetic field H_n (A/m)');
title('Normal H-field versus Relative Permeability');
% Verification of tangential H continuity and normal B continuity
fprintf('\nBoundary verification for zero surface current:\n');
fprintf('H1t = %.6f A/m, H2t = %.6f A/m\n',H1t,H2t);
fprintf('B1n = %.6e T, B2n = %.6e T\n',B1n,B2n)
```

OUTPUT:

```
--- Magnetic Boundary Condition Results ---
H2t (theoretical) = 10.000000 A/m
B2n (theoretical) = 2.000000e-03 T
H2n (theoretical) = 530.516477 A/m
Calculated mur2 = 3.183099
Reference mur2 = 3.000000
Percentage deviation = 6.1033 %
```



Test Case (ii) – Field Components and Refraction Angle at the Interface of Two Magnetic Media

Determine the field components, normal flux density and refraction angle in medium 2 for a specified incident field and specified relative permeabilities.

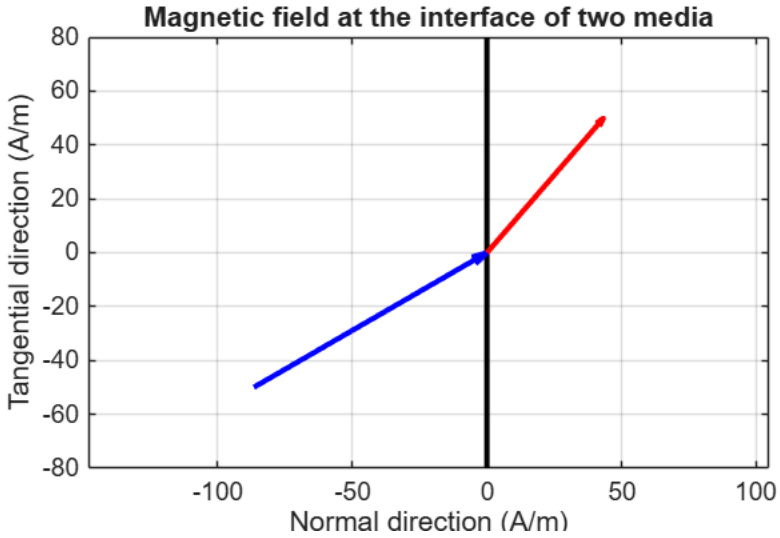
MATLAB Program:

```
clc; clear;
mu0 = 4*pi*1e-7;
```

```
H1 = input('Enter magnitude of H1 in medium 1 (A/m): ');
th1 = input('Enter angle of H1 with the normal (deg): ');
mur1 = input('Enter relative permeability of medium 1: ');
mur2 = input('Enter relative permeability of medium 2: ');
H1n = H1*cosd(th1); % normal component in medium 1
H1t = H1*sind(th1); % tangential component in medium 1
B1n = mu0*mur1*H1n; % normal flux density in medium 1
H2t = H1t; % tangential H is continuous (Ks = 0)
B2n = B1n; % normal B is continuous
H2n = B2n/(mu0*mur2); % normal H in medium 2
H2 = sqrt(H2n^2 + H2t^2);
th2 = atand(H2t/H2n);
fprintf('H1n = %.2f A/m, H1t = %.2f A/m\n', H1n, H1t);
fprintf('B1n = %.4e T\n', B1n);
fprintf('H2t = %.2f A/m, B2n = %.4e T\n', H2t, B2n);
fprintf('H2n = %.2f A/m, |H2| = %.2f A/m\n', H2n, H2);
fprintf('Refraction angle theta2 = %.2f deg\n', th2);
figure; plot([0 0],[-80 80],'k','LineWidth',2); hold on;
quiver(-H1n,-H1t,H1n,H1t,0,'b','LineWidth',2);
quiver(0,0,H2n,H2t,0,'r','LineWidth',2);
axis equal; grid on;
xlabel('Normal direction (A/m)'); ylabel('Tangential direction (A/m)');
title('Magnetic field at the interface of two media');
```

OUTPUT:

```
H1n = 86.60 A/m, H1t = 50.00 A/m
B1n = 1.0883e-04 T
H2t = 50.00 A/m, B2n = 1.0883e-04 T
H2n = 43.30 A/m, |H2| = 66.14 A/m
Refraction angle theta2 = 49.11 deg
```



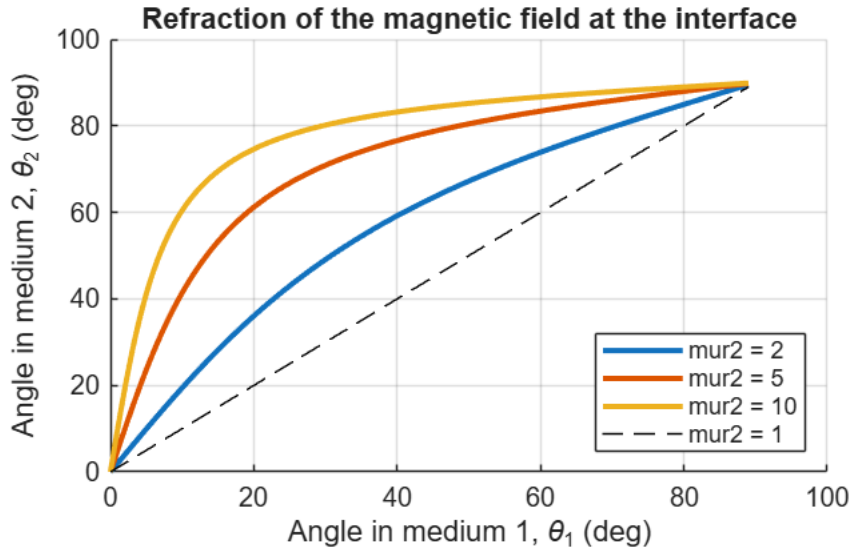
Test Case (iii) – Variation of the Refraction Angle with the Angle of Incidence for Different Permeabilities

Plot the angle of the magnetic field in medium 2 against its angle in medium 1 for three values of μr2 with μr1 = 1.

MATLAB Program:

```
clc; clear;
th1 = linspace(0,89,179);
mur1 = 1;
mur2 = [2 5 10];
figure; hold on;
for k = 1:3
th2 = atand((mur2(k)/mur1)*tand(th1)); % tan(th2) = (mur2/mur1)*tan(th1)
plot(th1,th2,'LineWidth',2);
end
plot(th1,th1,'k--'); % mur2 = mur1 (no change of direction)
grid on;
xlabel('Angle in medium 1, \theta_1 (deg)');
ylabel('Angle in medium 2, \theta_2 (deg)');
legend('mur2 = 2','mur2 = 5','mur2 = 10','mur2 = 1','Location','southeast');
title('Refraction of the magnetic field at the interface');
```

OUTPUT:



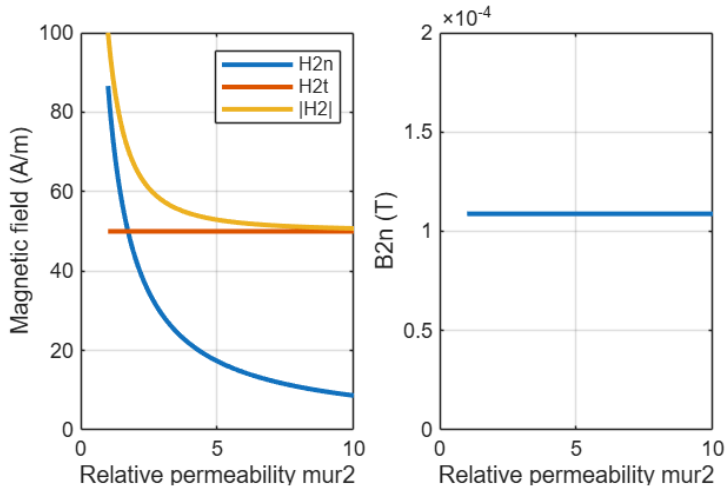
Test Case (iv) – Effect of the Relative Permeability on the Field Components in Medium 2

Keep the incident field fixed and vary μr2 to show which components are continuous across the boundary.

MATLAB Program:

```
clc; clear;
mu0 = 4*pi*1e-7;
H1 = 100; th1 = 30; mur1 = 1;
H1n = H1*cosd(th1); H1t = H1*sind(th1);
B1n = mu0*mur1*H1n;
mur2 = linspace(1,10,100);
H2n = B1n./(mu0*mur2); % normal H in medium 2
H2t = H1t*ones(size(mur2)); % tangential H (continuous)
H2 = sqrt(H2n.^2 + H2t.^2);
B2n = mu0*mur2.*H2n; % normal B (continuous)
subplot(1,2,1);
plot(mur2,H2n,mur2,H2t,mur2,H2,'LineWidth',2); grid on;
xlabel('Relative permeability mur2'); ylabel('Magnetic field (A/m)');
legend('H2n','H2t','|H2|');
subplot(1,2,2);
plot(mur2,B2n,'LineWidth',2); grid on; ylim([0 2e-4]);
xlabel('Relative permeability mur2'); ylabel('B2n (T)');
```

OUTPUT:



Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting	Marks given	Marks obtained
Identify magnetic media and boundary orientation	Identify magnetic media and boundary orientation	Correctly identifies permeability, normal/tangential components and surface-current assumptions.	Setup is correct with minor omissions.	Needs help identifying components or assumptions.	Uses incorrect interface definitions.	25	2.5	
Apply magnetic boundary relations	Apply magnetic boundary relations	Correctly applies continuity of normal B and the tangential H relation.	Relations are applied with minor errors.	Partly correct application requires guidance.	Boundary relations are incorrect.	25	2.5	
Determine permeability and validate fields	Determine permeability and validate fields	Calculates required quantities and verifies the boundary relations consistently.	Results are mostly correct with minor discrepancies.	Validation is incomplete.	Results are invalid or unverified.	30	3.0	
Explain and document findings	Explain and document findings	Explains magnetic-field behaviour and records assumptions, units and conclusion.	Documentation is clear with minor omissions.	Interpretation or documentation is incomplete.	Cannot explain or report results.	20	2.0	
TOTAL						100	10	